

Direct cross section measurement of $^{102}\text{Pd}(\gamma, p)$ and $^{102}\text{Pd}(\gamma, \alpha)$ for the astrophysical p process

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Background: A handful of neutron-deficient stable nuclei, known as the “ p nuclei,” cannot be produced through astrophysical neutron capture processes. Instead, some of these nuclei are proposed to be produced by γ -induced reactions on existing r - and s -process seeds. The specific astrophysical site or sites are not yet identified, however, with uncertainties in the cross sections of these γ -induced reactions playing a role. Databases of reaction rates for astrophysical simulations often rely on theoretical statistical model calculations, such as Hauser-Feshbach, for rates where no experimental information is known. However, reasonable variations in the choice of parametrizations of various nuclear properties can create order-of-magnitude variations in the final predicted cross sections and reaction rates, which are then propagated through the models to the predicted final abundances.

Purpose: To better constrain these statistical model calculations and ultimately reduce the uncertainties from the nuclear physics on our understanding of the p nuclei, a measurement of the cross sections of γ -induced reactions on the p -nucleus ^{102}Pd was undertaken. This work represents the first measurement of its kind, using segmented silicon detectors to measure prompt charged particle emission from γ -induced reactions.

Methods: Quasimonoeenergetic gamma beams from the High Intensity γ Source facility bombarded an enriched ^{102}Pd target. A segmented silicon array was arranged to detect the particles emitted from (γ, p) and (γ, α) reactions.

Results: Reaction cross sections were deduced at multiple γ -beam energies between 10 and 19 MeV, and compared to statistical model calculations using TALYS-1.96. The $^{102}\text{Pd}(\gamma, p)^{101}\text{Rh}$ reaction cross section was reasonably well reproduced by a subset of photon strength functions and level densities, though the strength to

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the ground state of ^{101}Rh was underestimated at higher incident gamma energies. The $^{102}\text{Pd}(\gamma, \alpha)^{98}\text{Ru}$ was in general overpredicted by the various alpha-nucleus optical model potentials.

Conclusions: While the theoretical cross sections used to model the (γ, p) reactions for the p process may be reasonable, a more careful approach is needed in the case of (γ, α) . Further work to probe gamma-induced reaction cross sections at and near the p nuclei is warranted.

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I. INTRODUCTION

As far back as the groundbreaking works of Burbidge *et al.* [1] and Cameron [2], the isotopes of the heavy elements present in the solar system have been separated into three groups, based on their nuclear properties and proposed astrophysical origin. The majority of these isotopes are formed by neutron capture, through either the slow neutron capture (s) process or the rapid neutron capture (r) process, but 35 heavy ($Z \geq 34$, $70 \leq A \leq 200$), neutron-deficient isotopes cannot be produced by either the r or s processes. These isotopes, identified as the “p nuclei,” are significantly rarer than their r- and s-process neighbors. There is therefore no element for which p nuclei dominate the stellar abundance, with molybdenum having the largest p-nuclei fraction at 24% in the Solar System (see, e.g., Refs. [3,4]). As a result, observations of Galactic Chemical Evolution have shed little light thus far on the site or sites of the p process (e.g. Refs. [4,5]). The story is complicated by some isotopes whose origin has traditionally been attributed to the p process that have potentially significant contributions from other processes: for example, ^{152}Gd and ^{164}Er from the s process, ^{113}In and ^{115}Sn from post-r-process decay chains, or ^{138}La and $^{180\text{m}}\text{Ta}$ from the ν process in core-collapse supernovae [6,7]. Further, no direct observations of p-nucleus production *in situ* have been made. Nonetheless, understanding the origins of the p nuclei is important for reproducing their abundances in galactic chemical evolution studies and indirect observations (e.g. Refs. [8,9]), and is even relevant to cosmochronology [10].

The consensus from nuclear physics, meteoritic data, and the modeling of stellar life cycles (see, e.g., Refs. [5,11]) is that there are at least two processes responsible for p-nuclei production. For the lighter p nuclei (perhaps $A < 100$), the νp process is thought to be the dominant source. Fröhlich *et al.* [12], Pruet *et al.* [13], and Wanajo [14] revealed a significant neutrino-driven flow to proton-rich nuclei above $A = 64$ in the range of $\approx 1\text{--}3$ GK, introducing the νp process and suggesting that the inner ejecta of CCSN could be the production site of the light p nuclei. Current multidimensional modeling [15–21] suggests that the lightest p nuclei, from ^{74}Se to ^{98}Ru , are produced within the hot, neutrino-heated bubble above the proto-neutron star in core-collapse and electron-capture supernovae.

However, the νp process in these events does not seem to be strong enough to reach above $A \approx 100$. It is therefore predicted that the heavier p nuclei are synthesized by photodisintegration of preexisting s- and r-processes nuclei via the (γ, n) , (γ, p) , and (γ, α) reactions at $\approx 2\text{--}3\text{-GK}$ temperatures in explosive astrophysical scenarios [5,22–27]. Various sites have been proposed for this so-called γ process, including

core collapse supernovae with helium- and carbon-burning progenitors [28], type II supernovae [24], and type Ia (thermonuclear) supernovae [29].

To predict p-nuclei abundances resulting from the γ process, accurate knowledge of the capture and photodisintegration reaction rates for thousands of stable and neutron-deficient nuclei is necessary, but modeling this nucleosynthesis process has thus far encountered difficulty in reproducing the observed abundances of the p nuclei in the solar system to better than a factor of ≈ 3 [29–31]. The astrophysical photon distribution in the thermodynamic conditions relevant for the γ process is broad, such that reaction cross sections must be known or predicted from sub-MeV up to tens of MeV. Significant efforts in the measurement of the gamma-induced reactions involved in the γ process have been made using various techniques [32–34], but (γ, p) and (γ, α) photodisintegration reaction cross sections have only infrequently been reported (see, e.g., Refs. [35,36]). Generally, one of two techniques is utilized: either the gamma-induced reactions are measured using bremsstrahlung radiation, or an indirect method or measurement of the time-inverse reaction is utilized instead.

In the first case, targets of interest are irradiated using bremsstrahlung photons, and the particle emission measured *in situ*, e.g., Ref. [37], or the delayed gammas from activation measured offline, e.g., Ref. [38]. The use of bremsstrahlung gammas requires that the resultant activation cross sections be carefully deconvolved from the incident photon distribution. Measurements of relative cross sections as a function of bremsstrahlung end-point energy or flux-weighted average energy avoid this difficulty, but conversion to absolute cross sections often relies on comparison to previously published data, and discrepancies exist [39]. Recent work in developing quasimonoeenergetic gamma beams for these types of studies has been undertaken [40–43], as gamma-induced reactions are relevant not only for astrophysics, but for many different applications [44].

Alternatively, indirect measurements can be used, including measurement of the time-inverse reactions, e.g. Refs. [45–68]. In some cases, measurements can be undertaken with stable beams; in a few recent measurements probing the astrophysical γ process, radioactive ion beams were utilized [50,51]. This technique is particularly powerful when it is anticipated that contributions from thermally excited states may play a large role in the stellar reaction rate. Photodisintegration reactions can provide crucial nuclear physics information for the astrophysical γ process when the ground-state contributions to the stellar rate are significant [69,70], as in the forward direction, thermally excited states

generally are not populated in the target or beam, and their contribution must be calculated separately. Conversely, the contribution to the reaction cross section due to excited final states is not captured in time-inverse reactions, and must be estimated as well. Hence, nuclear structure information in both the forward and inverse directions is crucial to reducing the uncertainties in the astrophysically relevant cross section.

To evaluate the reaction rates for which experimental cross section data are not available, state-of-the-art nuclear reaction models with up-to-date nuclear structure knowledge should be taken into account [71–74]. These theoretical estimates, however, remain uncertain due to the lack of complete nuclear structure knowledge, for example the nuclear optical model potential for protons and α particles at the energy region far below the Coulomb barrier [68,75]. Predictions from the commonly used statistical Hauser-Feshbach model, such as adopted in the code TALYS [76], can vary by orders of magnitude depending on the nuclear information used as inputs to the calculation, without some amount of experimental constraint. Nonstatistical processes may also play a role in the γ -induced cross sections [77]. Measurements of the photonuclear reactions (γ, p) and (γ, α) [78,79], in the mass and energy range of astrophysics interest, can thus bring new experimental constraint on the γ process specifically and on the nucleosynthesis of p nuclei more broadly, and inform the theoretical models used to calculate cross sections farther from stability. In particular, measurements at precise energies allow for detailed spectroscopy at each incident energy, which serves to probe the final state population of the photodissociation and constrain any inverse-reaction data such as (p, γ).

In the current work, a spectroscopic measurement of gamma-induced charged particle emission on the p-nucleus isotope ^{102}Pd was performed, as a step toward addressing the uncertainties in the nuclear physics of the astrophysical γ and p processes. This work constitutes the first time such a spectroscopic and kinematically complete in-beam measurement of two-body nuclear reactions generated by an incident gamma beam on a p nucleus has been reported. ^{102}Pd can be taken as representative of the uncertainties in photoinduced reactions on p nuclei in the region; additionally, tests of certain aspects of theoretical statistical models, such as the alpha-nucleus optical model potential, are more sensitive in the region of $A \approx 100$ [48,80].

Rauscher [81] identified ^{102}Pd as being “Priority I” for experimental study, as this isotope of Pd may represent the branching point where alpha emission exceeds other particle emission for temperatures relevant to the γ process (see also Ref. [54]). Crucially, as the lowest-mass p-process nucleus above mass 100, ^{102}Pd represents a potential transition point between the various proposed p-nuclei production mechanisms. Accurate knowledge of the reaction cross sections involved in the production and destruction of ^{102}Pd can thus assist in determining the contributions to its abundance from the various processes in an astrophysical environment.

II. EXPERIMENT

The measurement was performed at the High Intensity Gamma Source (HI γ S) [82] at the Triangle Universities

Nuclear Laboratory on the campus of Duke University. HI γ S utilizes a free electron laser and electron storage ring to generate quasimonoenergetic gamma beams ranging from 1 to 100 MeV through Compton backscattering. For this measurement, the gamma beams were circularly polarized and varied from 10 to 19 MeV, with a $\approx 3\%$ full width at half maximum (FWHM) energy spread, and delivered to the Upstream Target Room (UTR) vault. The setup is shown schematically in Fig. 1.

Before entering the UTR, the gamma beam passed through several stages of collimation, most importantly a beam-defining 10.5-mm-diameter lead collimator upstream and another 12-mm-diameter lead collimator downstream to remove edge-scattered gammas. This collimation ensured that the gamma beam impinged only target material, and not the thin aluminum target frames. Downstream of the collimators was a cinderblock wall and lead brick shielding, to ensure that no scattered gammas reached the target location. To reduce gamma-induced electrons reaching the target location, a long (≈ 2 m, ≈ 2.5 -cm diameter) vacuum pipe was placed through the port in the cinderblock wall between the collimators and the UTR, where it connected to the target chamber. The entrance port of the vacuum pipe was a blank aluminum flange that had been milled down to a wall thickness of approximately 1.3 mm (0.050 in.). By placing this entrance port far away from the target chamber and upstream of the cinderblock and lead shielding, the electron background in the detectors was significantly reduced.

The ^{102}Pd target was provided by Oak Ridge National Laboratory Stable Isotopes, and was known to be 78.18% enriched from assay by spark source mass spectrometry of the original source material [83]. A natural palladium target, composed of approximately 1% ^{102}Pd , 11% ^{104}Pd , 22% ^{105}Pd , 27% ^{106}Pd , 26% ^{108}Pd , and 12% ^{110}Pd , was measured to account for the cross section due to other isotopes of Pd. Additionally, natural zinc and copper were assayed in the enriched Pd target at the level of 1000 and 2000 ppm respectively, and thus were also measured separately to account for potential backgrounds. The target thicknesses were independently measured using a dedicated energy loss setup at the Center for Accelerator Target Science (CATS) at Argonne National Laboratory. This system utilizes the known stopping powers of alpha particles from a ^{244}Cm source in elemental materials to determine areal densities, using various levels of collimation, with an assigned uncertainty of 10% based on potential variations in thickness across the target area due to the fabrication process. The measured target thicknesses are listed in Table I.

To measure the charged particles emitted from the target, the Oak Ridge National Laboratory Silicon Detector ARray (SIDAR) was utilized. SIDAR is comprised of Micron Semiconductor YY1 style annular detectors in a “lampshade” configuration of six detectors. Two “lampshades” were placed around the target location, one upstream and one downstream, inside of the target chamber, which was held at millitorr vacuum. The upstream array was arranged into $\Delta E - E$ telescopes of 65- μm -thick detectors backed by 1000- μm -thick detectors; similarly, the downstream array comprised 110- μm -thick ΔE detectors and 1000- μm E detectors. Each detector

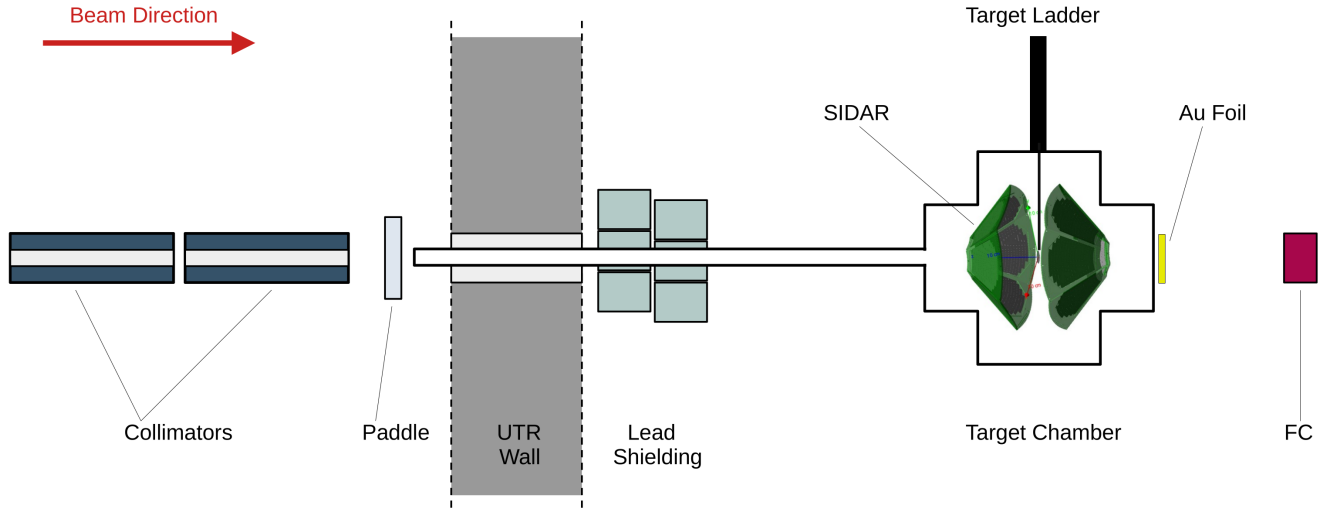


FIG. 1. A schematic of the experimental setup, not to scale. The gamma beam traverses left to right. The beam collimators and a thin scintillating plastic paddle are located upstream of the UTR. The target chamber, containing the detector array SIDAR, is located downstream of lead shielding, in the UTR. A gold foil and a fission chamber (FC) for beam normalization were mounted downstream of the target chamber. Further detail in the text.

has 16 annular strips, covering laboratory angles of $\approx 22^\circ$ – 61° (or $\approx 119^\circ$ – 158°) with roughly 2.5° per strip. The detectors were calibrated in energy and in geometric efficiency using a mixed triple-alpha source (^{239}Pu , ^{241}Am , ^{244}Cm) with known intensity placed at the target location. The geometric coverage of the detector arrays, 38.7%, was also calculated from the known detector positions relative to the target location, which agreed with the source measurements to within less than 1%. The experimental resolution was dominated by the target thickness. The standard ORNL analog data acquisition system [84], based on Mesytec MSCF-16 shaping amplifiers and CAEN V785 peak-sensing ADCs, was utilized for the collection of data.

Three separate techniques were implemented for determination of the absolute gamma flux on target. After the collimation but prior to entering the vacuum chamber, a scintillating paddle detector intercepted the gamma beam and gave an instantaneous measure of the beam flux. Downstream of the target, a gold foil was placed in the beam and activated for set periods of time. Each newly irradiated gold foil was then measured in a shielded counting station outside of the UTR and compared against the known $^{197}\text{Au}(\gamma, n)$ cross section [85]. Lastly, a small fission chamber (FC) [86] was placed at the end of the beam path in the UTR and used to get

a run-by-run measure of the γ beam intensity through the well-characterized cross section of γ -induced fission on ^{238}U [86–88]. Collimated beam intensities on target ranged from roughly 8×10^7 to 3×10^8 γ/s .

III. ANALYSIS

To identify the reaction products of interest (protons, $S_p = 7.809$ MeV, and alphas, $S_\alpha = 2.131$ MeV), two techniques were adopted. At most of the gamma beam energies measured, standard energy loss techniques could be used to identify particles based on the ratio of energies deposited in the ΔE and E detectors. For these measurements, a two-dimensional software gate was applied to separate protons and alphas from one another and from electronic and detector backgrounds. An example of this is shown in Fig. 2, with resultant proton

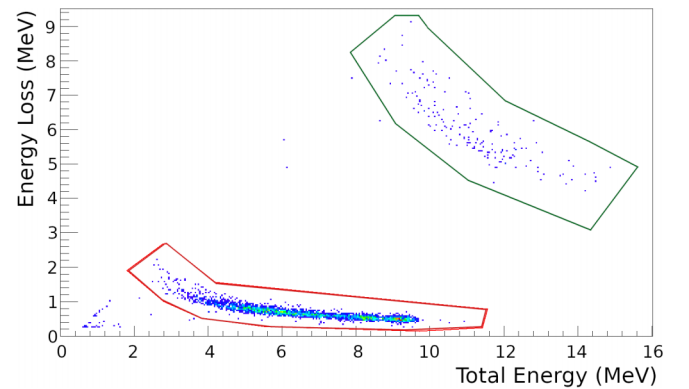


FIG. 2. A ΔE vs E_{tot} spectrum from one detector telescope, with the protons (red band) and alpha particles (green band) from gamma-induced reactions on the ^{102}Pd target identified. Data are shown for a 17-MeV gamma beam. Multiplicity and ray-tracing cuts have been applied; see text.

TABLE I. Targets used in the current measurement. All targets were self-supporting foils. Target areal densities are known to 10%.

Target	Areal density (mg/cm ²)	Atomic weight (amu)	Areal density (atoms/cm ²)
^{102}Pd	1.80	102.82	1.05×10^{19}
$^{\text{nat}}\text{Pd}$	1.38	106.42	7.79×10^{18}
$^{\text{nat}}\text{Cu}$	2.40	63.546	2.27×10^{19}
$^{\text{nat}}\text{Zn}$	1.09	65.38	1.00×10^{19}

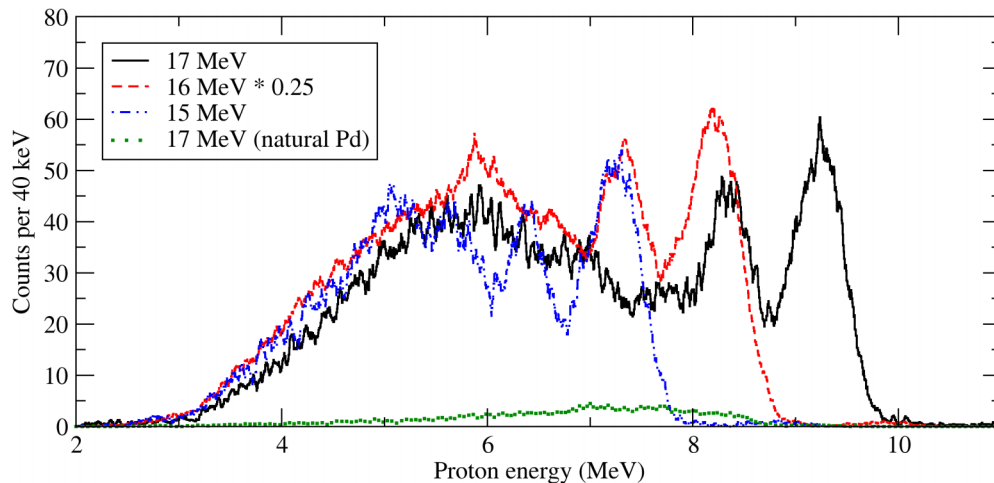


FIG. 3. Unnormalized energy spectra from the $^{102}\text{Pd}(\gamma, p)$ reaction, summed over all angles, for incident gamma energies of 17 MeV (black solid), 16 MeV (red dash), and 15 MeV (blue dot-dash). Note that the raw counts of the 16-MeV data have been normalized by a factor of 0.25 (indicated by $\times 0.25$ in the legend) to fall on the same plot. The 17-MeV energy spectrum for the natural Pd target (green dot) is also shown (normalized relative to the 17-MeV enriched Pd cross section) demonstrating that the structure apparent in the data originates from reactions on ^{102}Pd . The highest energy protons correspond to the lowest excitation energies in the final ^{101}Rh nucleus.

energy spectra shown in Fig. 3. A ray-tracing cut was also applied, ensuring that only particles which struck within one annular strip in the E detector compared with the ΔE were counted, reducing accidental coincidences. A multiplicity 2 cut was also applied, ensuring that no accidental hits of neighboring strips were summed in the total energy of an event. This technique allowed for clean separation of events for all reaction ejectiles with sufficient energy to punch through the ΔE layer of detectors (either 65 or 110 μm). No proton or alpha events had sufficient energy to punch through both detectors in the ΔE -E stack.

In the case of ^{102}Pd , the maximum proton energy from the (γ, p) reaction for the 19-MeV gamma beam is 11.22 MeV; at the 10-MeV gamma beam energy, this reduces to just over 2 MeV. The ejectiles from the (γ, α) reaction carry more energy due to the different separation Q value, but also have a higher stopping power. As such, standard energy loss techniques were not feasible for either reaction at the lowest gamma beam energies and a different particle identification technique had to be implemented.

The main background in the SIDAR detectors at these low particle energies are delta electrons created by scattered gammas from the beam striking materials in and around the target chamber. Because the stopping power for electrons is small compared to the heavier reaction ejectiles, these particles leave very little energy in the detectors, creating a smooth background that extends into the region where the lowest energy protons and alphas are located. Because these electrons do not stop in either detector of the telescope, however, the E detector can be used to veto many of these background events. Spectra were generated where events with a signal in the E detector were vetoed, leaving only events which deposited energy in the ΔE detector; this removed electron background above roughly 1.3 MeV in the 110- μm downstream detectors, and roughly 750 keV in the upstream 65- μm detectors. The known reaction kinematics were then used to determine

regions of interest (ROI) in the particle energy for the ejectile protons and alphas, demonstrated in the spectrum in Fig. 4, and these regions integrated. The regions were intentionally wide to account for the broadening effects of the target thickness, ejectile angle, and detector dead layers. A systematic uncertainty due to the potential unvetted background signals from the delta electrons was included in addition to the statistical uncertainty.

This E-vetoed-event technique was applied to the data at all energies, and the resulting count of protons and alphas was summed with the events previously identified through standard PID as described above. No double counting of events could occur due to the E detector veto requirement. Above 12-MeV beam energy, the number of proton events identified outside of the standard ΔE -E dropped below 20%, falling to under 1% at the highest energies. For the (γ, α) reaction channel, the fraction of events identified using the veto remained significant at all beam energies.

As the ^{102}Pd target contained a substantial fraction of the other Pd isotopes, a natural Pd target was also measured at most of the incident gamma energies. At $E_\gamma = 10, 16$, and 18 MeV, the natural Pd target was not measured. At 10 MeV, the (γ, p) and (γ, α) cross sections on natural Pd were predicted to be too low to reliably measure; at the two higher energies, the cross section varies smoothly and is straightforward to estimate. For these energies, the contribution to the enriched target cross section was estimated using a linear interpolation (or extrapolation, in the case of the 10-MeV point) of the $^{\text{nat}}\text{Pd}$ yield. While the cross section is not linearly changing, using a linear interpolation over these small energy steps provided a simple way to estimate the yield without requiring normalization to theoretical cross section calculations. This technique is further justified by the difference in cross sections for the ^{102}Pd compared with the other stable isotopes of palladium, as seen in Fig. 5; in the case of both (γ, p) and (γ, α) , the cross section on ^{102}Pd is almost an order of magnitude higher than

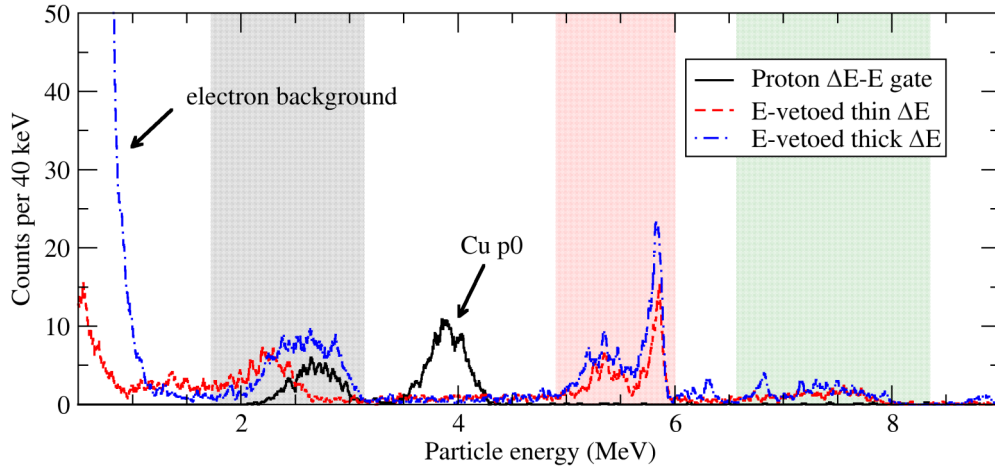


FIG. 4. Energy spectra from γ -induced reactions on the ^{102}Pd target at an incident gamma energy of 10 MeV, summed over all angles, using the E-vetoed- ΔE technique (see text). The spectrum obtained through the standard ΔE -E techniques for this same gamma energy is shown in black. The two different thicknesses of ΔE layer silicon produced the red-dash and blue-dot-dash spectra; the thicker the detector, the higher energy the beam-induced background. A structure arising from the (γ, p) reaction on the copper contaminant is labeled; see text. The proton (gray shading) and alpha (green shading) regions of interest are shown. Also shown is a region (red shading) around the peaks at 5.5–6 MeV, which are residual weak alpha activity in the vacuum chamber.

the next strongest isotope (^{104}Pd). Hence an uncertainty in the natural palladium cross section, as well as the uncertainty due to the differences in the ratios of the Pd isotopes other than $A = 102$ in the natural versus the enriched targets, can be anticipated to have a small effect on the extracted ^{102}Pd cross section from the enriched target. A systematic uncertainty accounting for this interpolation was included in the subtracted $^{\text{nat}}\text{Pd}$ cross sections.

Due to their lighter mass, the predicted cross sections on the copper and zinc contaminants in the enriched ^{102}Pd target are greatly enhanced at the lower gamma beam energies relative to the cross sections for Pd. The peak corresponding to $^{65}\text{Cu}(\gamma, p0)$ falls just above the kinematically accessible region of (γ, p) on the palladium isotopes (see Fig. 4), but (γ, p) to higher-lying excitation energies in ^{64}Ni , as well as

the (γ, α) , fall in the ROI for the events of interest on Pd. This is also true for reactions on the other isotope of copper, ^{63}Cu , and reactions on the various stable isotopes of zinc. To account for this background, the gamma-induced reaction cross sections on natural Cu and Zn targets (Table I) were measured separately for the same incident beam energies, utilizing the same experimental setup and analysis, and these cross sections subtracted from the enriched Pd target measurements. At the lowest energies, reactions on $^{\text{nat}}\text{Cu}$ corresponded to nearly half of the total yield from the enriched Pd target, emphasizing the importance of knowing the target assay.

Finally, to convert the experimental proton and alpha yields to absolute cross section, the run-by-run gamma beam flux as measured by the downstream FC [86,88] was used. As a consistency check, the absolute beam flux as determined from the FC was compared to the relative beam intensity from the upstream scintillator paddle; this is displayed in Fig. 6. Uncertainties on the FC gamma flux are on the order of 3.5–4% [86,88]. The paddle is not used for an absolute measure of the gamma beam intensity, but the instantaneous rate (γ/s) from the paddle was recorded as a scaler in the data stream, and the total number of gammas observed in the paddle calculated run by run based on the data acquisition start and stop timestamps. In the energy range measured here, which includes all runs on natural Pd, natural Cu, natural Zn, and enriched ^{102}Pd , the paddle response was highly linear, as can be seen in Fig. 6, allowing it to be calibrated to the actual gamma flux with a variance of $< 7\%$. While it can be anticipated that the scintillator paddle has an energy-dependent response, no trends were observed in the behavior of the paddle-FC correlation as a function of incident gamma beam energy, demonstrating that the paddle was operating in a stable, linear regime far from instrument thresholds. This 7% uncertainty was included in the final cross section error budget.

In addition to the FC and paddle, the gamma beam intensity was measured offline using activation of gold foils.

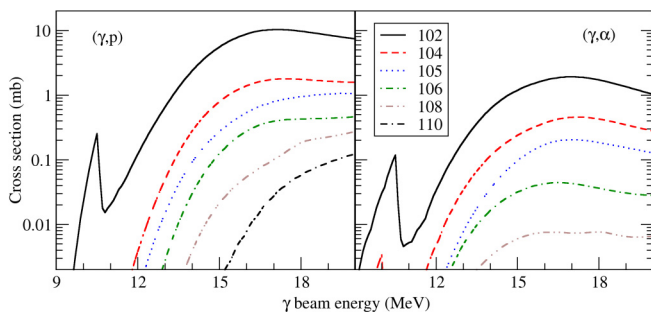


FIG. 5. Calculated cross sections for (γ, p) , left, and (γ, α) , right, on the naturally occurring palladium isotopes, using the default flags in TALYS-1.96 (see Sec. IV). The p-nucleus ^{102}Pd is shown in solid black, with isotopes 104 (red dash), 105 (blue dot), 106 (green dot-dash), 108 (brown dot-dot-dash), and 110 (black dash-dash-dot) falling at lower values. The structure seen most clearly in the ^{102}Pd spectrum is due to the opening of the neutron emission channel at 10.572 MeV.

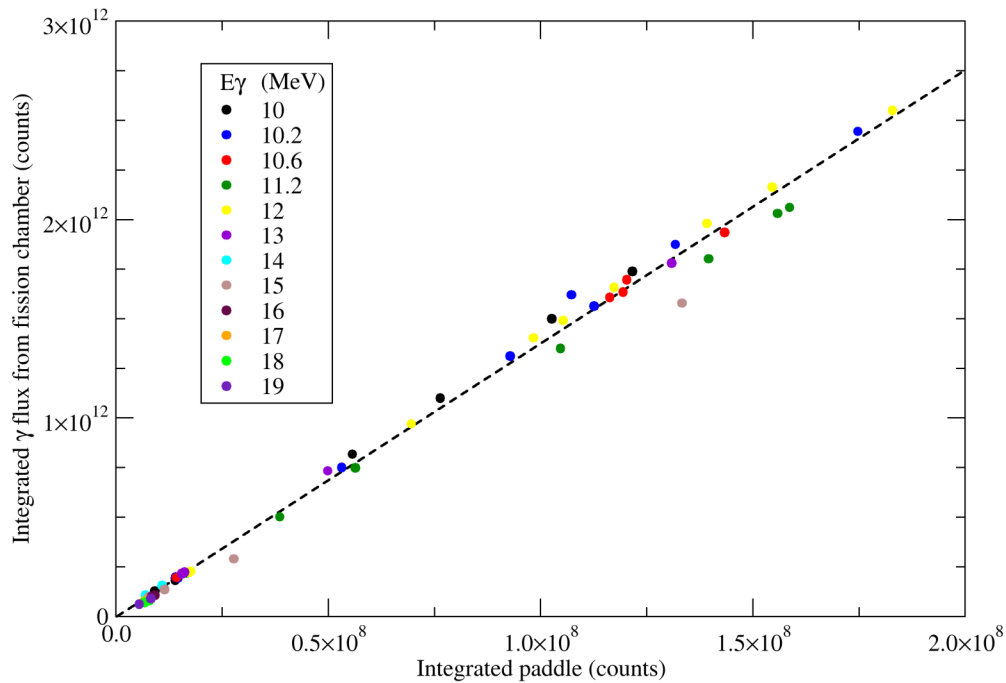


FIG. 6. Integrated gamma flux from the downstream FC compared to the integrated gamma flux as measured by the upstream scintillator paddle. Each point represents one data run, with each beam energy setting (ranging from 10 to 19 MeV, shown in different colors) comprising one or more runs. Runs were on the order of 1–2 h each, though the short (≈ 10 min) runs on the background targets are also included here. The black dashed line is a linear fit to the data, demonstrating that the relative measurement of the paddle is an accurate proxy for the actual gamma flux as measured by the FC.

The foils were irradiated over the course of one or more runs (for a single gamma beam energy at a time). The signature transitions from the decay of the ^{196}Au ground state through either ^{196}Pt (333.03, 355.73 keV) or ^{196}Hg (426.10 keV) originating from the newly irradiated foils were measured in a shielded HPGe counting station outside of the UTR [85], and compared to the measured $^{197}\text{Au}(\gamma, n)^{196}\text{Au}$ cross sections of Veyssiere *et al.* [89]. This method provided an external verification of the average gamma flux compared with the FC and the instantaneous rate from the paddle. The gamma fluxes obtained considering the $^{197}\text{Au}(\gamma, n)^{196}\text{Au}$ cross section in this manner agreed with those of the FC to within $\approx 2\%$. However, additional literature data for this cross section (e.g. Refs. [87,90]) show discrepancies above about 16 MeV, and the uncertainty derived from the gold activation at these higher incident gamma beam energies is slightly higher, roughly 10–12%, than the 7% from the FC. [A reevaluation of the $^{197}\text{Au}(\gamma, n)^{196}\text{Au}$ cross section is in preparation separately, based on Ref. [85].] Hence, for the current data, the FC flux was used to determine the cross section on Pd for runs at gamma energies below 16 MeV. For energies of 16 MeV and above on Pd where the FC was not instrumented, the calibrated paddle flux is used.

Isotropic distributions are assumed in the reaction product detection efficiency correction; this assumption is justified as the combination of the beam energy spread, the large angular and excitation energy range of the detected reaction products, and the relatively high level densities compared to the experimental resolution effectively samples a broad range of spins and parities in both the initial and final

states, washing out the impacts of individual anisotropies. The differing data acquisition live times, gamma fluxes, target thicknesses, and (in the case of Pd) isotopic abundances were accounted for in calculating the final ^{102}Pd and background target cross sections. The measured cross sections for $^{102}\text{Pd}(\gamma, p)^{101}\text{Rh}$ and $^{102}\text{Pd}(\gamma, \alpha)^{98}\text{Ru}$ are presented in Table II.

The $^{102}\text{Pd}(\gamma, p)^{101}\text{Rh}$ and $^{102}\text{Pd}(\gamma, \alpha)^{98}\text{Ru}$ cross sections as a function of incident gamma energy are shown in

TABLE II. Measured cross sections for the (γ, p) and (γ, α) reactions on ^{102}Pd . Uncertainties listed are 1σ , and include both statistical and systematic (backgrounds, beam flux normalization, target thickness, and detection efficiency) components, as described in the text.

E_γ (MeV)	$^{102}\text{Pd}(\gamma, p)^{101}\text{Rh}$ (mb)	$^{102}\text{Pd}(\gamma, \alpha)^{98}\text{Ru}$ (mb)
10	0.05 ± 0.01	0.009 ± 0.003
10.2	0.11 ± 0.02	0.017 ± 0.005
10.6	0.09 ± 0.04	0.015 ± 0.004
11.2	0.07 ± 0.01	0.004 ± 0.001
12	0.15 ± 0.02	0.028 ± 0.005
13	0.64 ± 0.09	0.10 ± 0.01
14	1.61 ± 0.21	0.23 ± 0.03
15	4.44 ± 0.57	0.47 ± 0.06
16	9.00 ± 1.13	0.74 ± 0.09
17	15.48 ± 1.95	0.98 ± 0.12
18	16.24 ± 2.04	0.98 ± 0.12
19	15.21 ± 1.91	0.93 ± 0.12

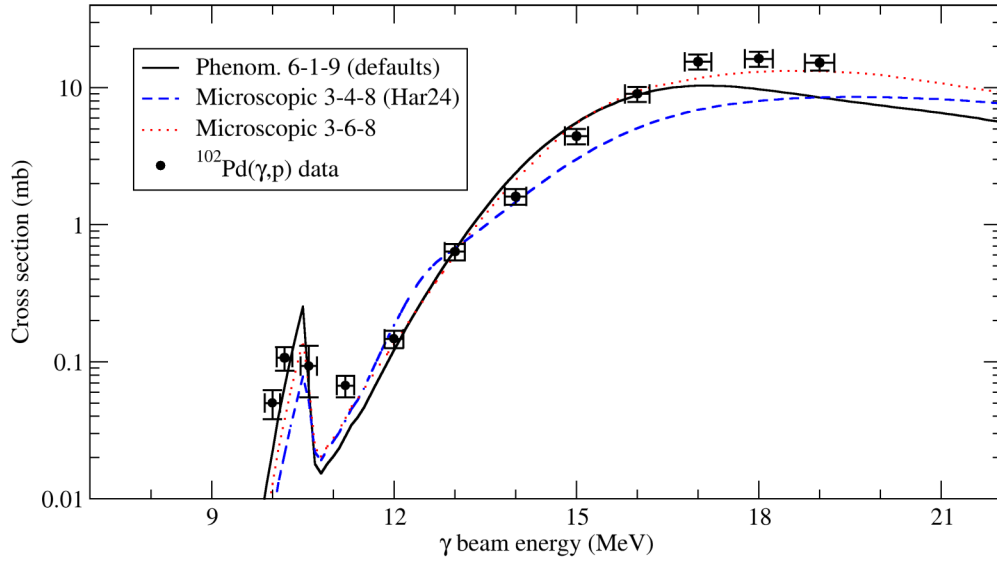


FIG. 7. Comparison between the extracted $^{102}\text{Pd}(\gamma, p)^{101}\text{Rh}$ cross section data from the current work (black circles) and a selection of physically motivated Hauser-Feshbach predictions calculated using TALYS-1.96, labeled by choice of *alphaomp-lmodel-strength* (see text). The default TALYS-1.96 output, which is purely phenomenological, is shown as a solid black line. The microscopic model combination recommended by Ref. [57] is labeled Har24 (blue dash). A similar combination but utilizing the temperature-dependent GHFB level density model is also shown (red dot).

Figs. 7 and 8, respectively. In both cases, the cross section uncertainties displayed include both statistical and systematic components (such as the beam and target density normalizations), as in Table II. The horizontal error bars represent the 3% FWHM energy spread of the gamma beam, as determined in previous experiments at the HI γ S facility [91,92] and verified during the current measurement.

IV. DISCUSSION

The absolute cross sections determined in the current work were compared against calculations using the Hauser-Feshbach formalism, specifically via the TALYS-1.96 software package [76]. Multiple calculations were performed, using the built-in nuclear physics parameters in TALYS: level densities (*lmodel*), gamma strength functions (also called photon

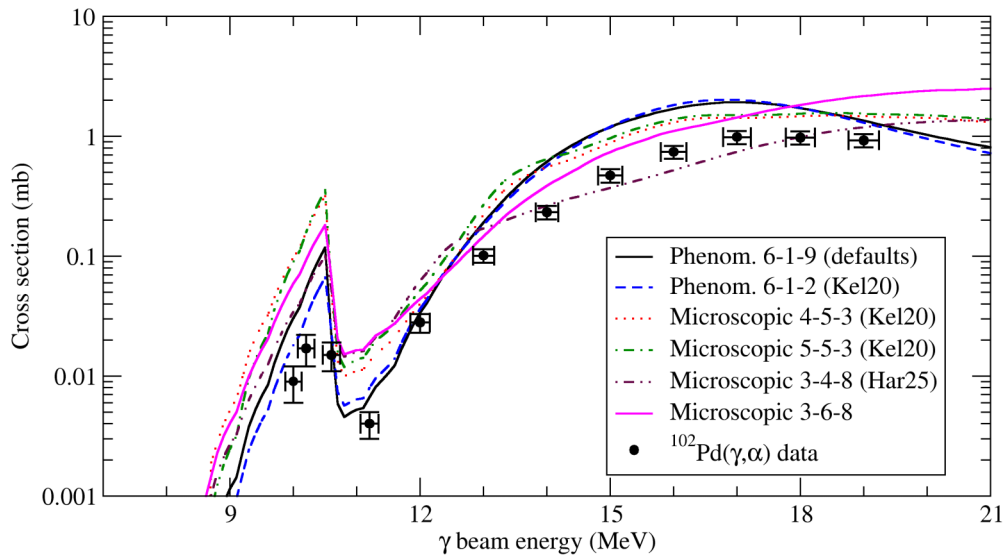


FIG. 8. Comparison between the extracted $^{102}\text{Pd}(\gamma, \alpha)^{98}\text{Ru}$ cross section data from the current work (black circles) and a selection of physically motivated Hauser-Feshbach predictions calculated using TALYS-1.96, labeled by choice of *alphaomp-lmodel-strength* (see text). The purely phenomenological predictions include the default TALYS-1.96 output (black solid) and a similar combination utilizing a Brink-Axel Lorentzian gamma strength function parametrization (blue dash). Two microscopic combinations recommended by Ref. [58] (Kel20) and one recommended by Ref. [48] (Har25) are also shown as red dot, green dot-dash, and purple dot-dot-dash. The temperature-dependent GHFB level density model combination shown to match the (γ, p) channel is also plotted here (magenta solid).

strength functions, *strength*), and the various nucleon-nucleus optical model potentials [76].

For (γ, p) , the proton-nucleus optical model potential is a spherical global model by default, based on the parametrization of Ref. [93]. Calculations using either the local or global parametrization (*localomp* y or n) resulted in identical (γ, p) cross section predictions. In the case of the alpha-nucleus optical model potential (*alphaomp*), multiple options are available: a “normal” potential (1), McFadden and Satchler [94] (2), microscopic potentials from Demetriou *et al.* [80] (3–5), the default potential from Avrigeanu *et al.* [95] (6), Nolte *et al.* [96] (7), and an older potential from Avrigeanu *et al.* [97] (8). It is well known that the potentials of Nolte *et al.* [96] and Avrigeanu *et al.* [97], which served as an extension to the parametrization of Ref. [96], are generally a poorer fit to low energy alpha-nucleus interactions.

The level density (*ldmodel*) options are identical for both the (γ, p) and (γ, α) channels, as are the (E1) gamma strength function (*strength*) options. The following level density models are built in to TALYS: the default constant temperature plus Fermi gas model (1), a (global) back-shifted Fermi gas model (2), a generalized superfluid model (3), and Skyrme-Hartree-Fock-Bogoluybov (4), Gogny-Hartree-Fock-Bogoluybov (5), or temperature-dependent Gogny-Hartree-Fock-Bogoluybov (6) level densities from numerical tables [76]. The built-in gamma strength functions in TALYS are a Kopecky-Uhl generalized Lorentzian (1), the default Brink-Axel Lorentzian (2), Hartree-Fock BCS tables (3), Hartree-Fock-Bogoliubov tables (4), a hybrid model from Ref. [98] (5), temperature-dependent Hartree-Fock-Bogoluybov (6) or relativistic mean field (7) models, a Gogny-Hartree-Fock-Bogoliubov QRPA model (8), and a default simplified modified Lorentzian (SMLO, 9). The microscopic level density and strength function models are not by default scaled to fit experimental data, and were also not scaled in the current work.

For $^{102}\text{Pd}(\gamma, p)^{101}\text{Rh}$ (Fig. 7), comparisons are made against the TALYS-1.96 default parameters, which are purely phenomenological, as well as a microscopic model combination recommended by Harissopulos *et al.* for (p, γ) reactions on the nearby $A \approx 110$ tin isotopes [57]. This microscopic model combination utilizes the first of the three Demetriou and Goriely alpha-nucleus optical model potentials (*alphaomp* 3), the Skyrme-Hartree-Fock-Bogoluybov level density combinatorial tables (*ldmodel* 4), and the temperature-dependent HFB + QRPA gamma strength function calculations using the D1M Gogny force (*strength* 8). Neither of these combinations was found to well reproduce the data. Recent work indicates the phenomenological level density models do not reproduce known level densities as accurately [99]. A purely microscopic combination using the recommended *alphaomp* and *strength* from Ref. [57] combined with the temperature-dependent Gogny-Hartree-Fock-Bogoluybov level density model (*ldmodel* 6) better reproduced the data, indicating that the temperature dependence of both the level density and strength functions is an important component to the statistical model predictions.

The possibility of two-proton emission was considered, as the (γ, p) data generally level off at higher energies, such that the TALYS predictions underestimate the data above about

16–17 MeV. In TALYS, the predicted 2p channel is more than a factor of 1000 weaker than the 1p channel; in fact, all two-particle channels are substantially weaker at these energies. Within the proton ΔE -E gate, the ratio of multiplicity 2 (one hit in each layer of the telescope) to multiplicity 3 or higher events was extracted. At the higher incident gamma energies, the number of higher multiplicity events represents under 5% of the total number of proton events. While the majority of these higher-multiplicity events are due to accidental coincidences, consistent with the data acquisition livetimes observed in this measurement, this 5% value can also be taken as an upper limit on the experimental 2p channel strength.

The measured $^{102}\text{Pd}(\gamma, \alpha)^{98}\text{Ru}$ reaction cross sections are shown compared to TALYS in Fig. 8. As with the (γ, p) channel, the data are compared to several recommended combinations of input parameters, including two purely phenomenological combinations and four microscopic combinations based on the recommendations of Harissopulos *et al.* [48] and Kelmar *et al.* [58], based on comparisons to measured $^{102}\text{Pd}(\alpha, \gamma)$ data. The microscopic model combination of Ref. [48] is identical to that recommended in Ref. [57]. The two microscopic model combinations of Ref. [58] include the two additional alpha-nucleus optical model potentials of Demetriou and Goriely (*alphaomp* 4,5), the Gogny-Hartree-Fock-Bogoluybov level density combinatorial tables (*ldmodel* 5), and the Hartree-Fock BCS tables for the gamma strength function (*strength* 3). Again, none of the predictions are a good match to the data, though the microscopic models tend to overpredict the lowest energy cross sections more than the phenomenological models. At the higher gamma energies, the two microscopic model predictions recommended by Ref. [58] overpredict the cross section but reasonably reproduce the overall shape of the cross section data, compared to the combination of Ref. [48].

As with the $^{102}\text{Pd}(\gamma, p)^{101}\text{Rh}$ data, a microscopic model combination utilizing temperature dependence in both the level density and gamma strength function is compared to the $^{102}\text{Pd}(\gamma, \alpha)^{98}\text{Ru}$ reaction cross sections. It is noteworthy that the temperature-dependent GHFB numerical tables perform reasonably well for both reaction channels (proton and alpha) simultaneously.

Importantly, in the case of the $^{102}\text{Pd}(\gamma, \alpha)^{98}\text{Ru}$ reaction, the alpha-nucleus optical model potential has the largest impact on the TALYS predictions, indicating that alpha scattering measurements on p nuclei—in particular around the neutron separation energy—can be of tremendous use in constraining the gamma-induced cross sections. This variability represents an order of magnitude at some energies; notably, as the majority of the *alphaomp* calculations considerably overpredict the measured cross sections, this variability cannot be taken as a realistic uncertainty in the statistical model predictions for the purposes of astrophysical modeling.

In the case of the $^{102}\text{Pd}(\gamma, p)^{101}\text{Rh}$ reaction, clear structure can be seen in the excitation energy spectra of the emitted protons (Fig. 3). The highest energy protons correspond to the ground state of ^{101}Rh . This high-energy peak is easily separable from the remaining structure with a standard Gaussian fitting procedure at all but the lowest gamma beam energies. Figure 9 shows the extracted $^{102}\text{Pd}(\gamma, p0)^{101}\text{Rh}$ partial cross

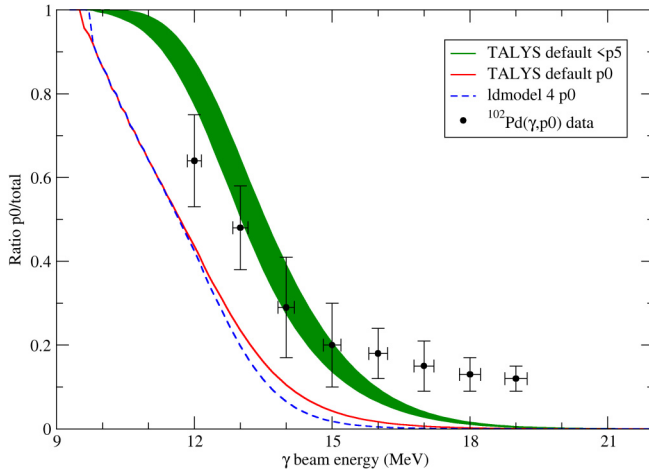


FIG. 9. Comparison between the extracted $^{102}\text{Pd}(\gamma, p0)^{101}\text{Rh}$ cross section data from the current work (black circles) and various predictions from TALYS. The ratio of the p0 (ground state) channel cross section to the total for (γ, p) was calculated for the TALYS default input options (red line) and for the *ldmodel* 4 (blue dash). Also shown is a calculation using the default level density of the first four (lower bound of green band) to five (upper bound of green band) levels in ^{101}Rh summed. See text.

section data from the current work, under the assumption that the area under the peak in each spectrum corresponds to the p0 channel, and compares it with various predictions from TALYS. The ratio of the p0 (ground state) channel cross section to the total for (γ, p) was calculated for the TALYS-1.96 default input options, and also for the *ldmodel* 4 [76], which was shown to have the largest impact on this ratio out of all level density models. Both calculations underestimate the data. However, as the experimental resolution was larger than the separation between the lowest-lying excitation energies in ^{98}Ru , a comparison with the ground state alone may not be sufficient. As such, the ratio of the TALYS cross sections to the first four to five levels was summed for comparison to the data. This encompasses the levels that fall within the FWHM of the experimental resolution (the first four levels being the lower limit, and the first five levels being the upper limit, shown as a green band in the figure). The agreement for this sum is much better than for p0 alone. Despite this general agreement, it can be seen that TALYS overpredicts at lower incident gamma energies and underpredicts at the higher energies; this is opposite to the expected behavior, given that the experimental resolution is slightly better for the higher-energy protons.

V. CONCLUSION AND FUTURE WORK

Understanding the production of p nuclei requires benchmarking of the theoretical statistical model predictions of the gamma-induced reaction cross sections involved. By demonstrating the first-ever gamma-induced charged particle measurement on a p nucleus at the HI γ S Facility, these cross sections can be measured experimentally and compared to the predictions from well-used statistical model codes such as TALYS [76]. In the case of ^{102}Pd , the (γ, p) and (γ, α) cross sections were measured across a range of incident gamma

energies, and compared to TALYS predictions based on the built-in nuclear level density, gamma strength function, and alpha-nucleus optical models.

Several of the built-in TALYS-1.96 options for the level density model and gamma strength function were found to reasonably reproduce the $^{102}\text{Pd}(\gamma, p)^{101}\text{Rh}$ reaction cross sections, though the strength to the p0 channel predicted by the statistical models showed an energy dependence that was not in the measured partial cross sections. In the case of $^{102}\text{Pd}(\gamma, \alpha)^{98}\text{Ru}$, the predicted cross sections generally overpredicted the measured cross sections; the theoretical prediction was particularly sensitive to the alpha-nucleus optical model parameters, none of which were found to be a good match to the data. The best match to the (γ, α) data is found to still overpredict the measured cross sections by up to a factor of ≈ 3 .

More measurements are needed to further constrain the (γ, α) reactions relevant to the production of p nuclei, including higher-statistics measurements of ^{102}Pd probing toward lower gamma beam energies. Measurements across isotopic chains relevant to γ -process nucleosynthesis of p nuclei will also help to elucidate the strengths and weaknesses of the various models incorporated into statistical codes such as TALYS.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available upon publication because it is not techni-

cally feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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